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QUESTION: 1

1) ~~find~~ $f(1, 3)$

Sol: ~~find~~ $f(x, y) = \frac{x^2 y}{2x - y^2}$; put values ; $f(1, 3) = \frac{1^2(3)}{2(1) - 3^2}$

$$f(1, 3) = -\frac{3}{7}$$

2) $f(-2, -1)$

Sol: $f(x, y) = \frac{x^2 y}{2x - y^2}$; put values ; $f(-2, -1) = \frac{(-2)^2(-1)}{2(-2) - (-1)^2}$

$$f(-2, -1) = \frac{4}{5}$$

3) $f(x+h, y)$

Sol: points $(x+h, y)$.

$$f(x+h, y) = \frac{(x+h)^2(y)}{2(x+h) - y^2}$$

4) $f(x, x)$

Sol: points (x, x) .

$$f(x, x) = \frac{x^2 x}{2(x) - x^2} = \frac{x^3}{x(2-x)}$$

$$f(x, x) = \frac{x}{2-x}$$

QUESTION : 3

$$g(x, y) = x^2 \ln(x+y)$$

Sol: $g(x, y) = x^2 \ln(x+y)$

a) $g(3, 1) = ?$

$$g(3, 1) = 3^2 \ln(3+1) = 9 \ln 4$$

b) domain of $g = ?$

Since domain of $\ln > 0$, so domain of $x^2 \ln(x+y)$ should be $x+y > 0$; $y > -x$.

$$\text{domain of } g(x, y) = \{(x, y) \mid y > -x\}$$

c) Range of $g = ?$

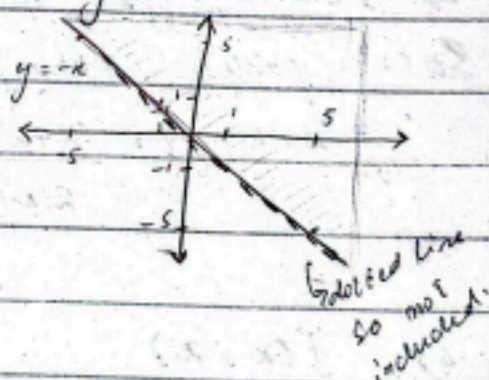
range of $\ln(x+y)$ is $\mathbb{R}$. so range of g is $\mathbb{R}$.

For graph of domain;

$$x = 1, y = -x \Rightarrow y = -1$$

$$x = 5, y = -5; x = -1 \Rightarrow y = 1$$

$$x = -5 \Rightarrow y = 5$$



QUESTION : 7

$$f(x, y) = \sqrt{x-2} + \sqrt{y-1}$$

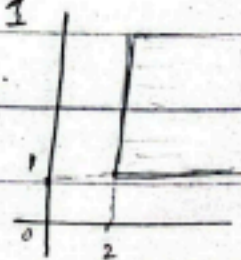
Solution:

$$f(x, y) = \sqrt{x-2} + \sqrt{y-1}$$

The domain of f is $x-2 \geq 0$ and $y-1 \geq 0$

$$\text{OR } x \geq 2 \text{ and } y \geq 1$$

So the domain is $\{(x, y) \mid x \geq 2, y \geq 1\}$.



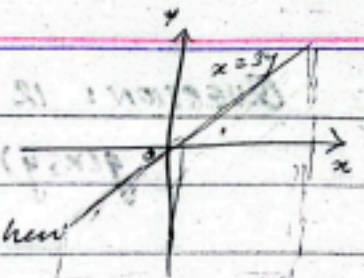
QUESTION: 8

$$f(x, y) = \sqrt{x-3y} \cdot \sqrt{x-3y}$$

Sol: The function f is defined when

$$x-3y \geq 0 \quad \text{OR} \quad x \geq 3y$$

So the domain of f is $\{(x, y) \mid x \geq 3y\}$.



QUESTION: 9

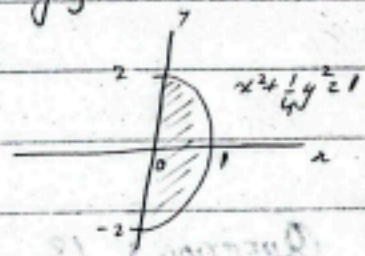
$$g(x, y) = \sqrt{x} + \sqrt{4-4x^2-y^2} \cdot \sqrt{x}$$

Sol: function g is defined when

$$x \geq 0 \quad \text{and} \quad 4-4x^2-y^2 \geq 0$$

$$\text{OR} \quad x \geq 0 \quad \text{and} \quad x^2 + \frac{1}{4}y^2 \leq 1$$

So the domain is $\{(x, y) \mid x^2 + \frac{1}{4}y^2 \leq 1, x \geq 0\}$.



QUESTION: 10

$$g(x, y) = \ln(x^2+y^2-9) \cdot \ln(x^2+y^2-9)$$

Sol: function g is defined when $x^2+y^2-9 > 0$

$$x^2+y^2 > 9 \quad \text{So the domain of } g \text{ is}$$

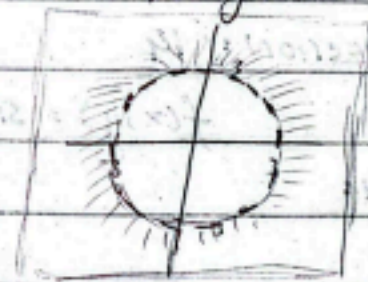
$$\{(x, y) \mid x^2+y^2 > 9\}$$

QUESTION: 11

$$g(x, y) = \frac{x-y}{x+y}$$

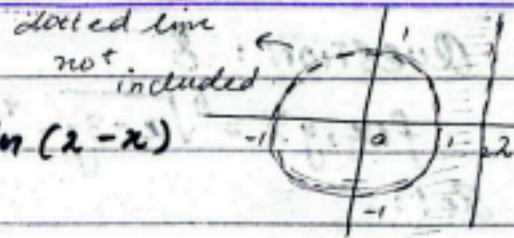
Sol: g is defined when $x+y \neq 0$ OR $y \neq -x$.

So domain of g is $\{(x, y) \mid y \neq -x\}$.



QUESTION: 12

$$g(x, y) = \frac{\ln(2-x)}{1-x^2-y^2} \cdot \ln(2-x)$$



Sol: The function g is defined only when:
 $2-x > 0$ and $1-x^2-y^2 \neq 0$

or $x < 2$ and $x^2+y^2 \neq 1$.

So domain of g is $\{(x, y) \mid x < 2, x^2+y^2 \neq 1\}$.

QUESTION: 13

$$p(x, y) = \frac{\sqrt{xy}}{x+1} \cdot \sqrt{xy}$$

Sol:

function p is defined when $xy \geq 0$ & $x+1 \neq 0$
or $x \neq -1$

So domain of p is:

$$\{(x, y) \mid xy \geq 0, x \neq -1\}$$

QUESTION: 14

$$f(x, y) = \sin^{-1}(x+y) \cdot \sin^{-1}(x+y)$$

Sol: The function f is defined only when

$$-1 \leq x+y \leq 1 \quad \text{OR} \quad -1-x \leq y \leq 1-x$$

So the domain of f is;

$$\{(x, y) \mid -1-x \leq y \leq 1-x\}$$

QUESTION: 17

$$S = f(w, h) = 0.0072 w^{0.425} h^{0.725}$$

Sol: a) $f(73, 178)$

$$f(w, h) = f(73, 178) = 0.0072 (73)^{0.425} (178)^{0.725}$$

$$f(73, 178) = 1.9$$

means a person weighing 73 kg and having a height of 178 cm has a surface area of 1.9 sq.m.

b) $f(54, 162.5)$

$$S = f(54, 162.5) = 0.0072 (54)^{0.425} (162.5)^{0.725}$$

$$S = 1.5$$

QUESTION: 33

a) $f(-3, 3)$: The point $f(-3, 3)$ lies between level curves 50 and 60 and it is much closer to 60 so we assume it to be at point 56.

b) $f(3, -2)$: The point $f(3, -2)$ lies half-way between curves 30 and 40 so we assume the point to be at 35.

QUESTION : 45

$$f(x, y) = x^2 - y^2$$

Sol:

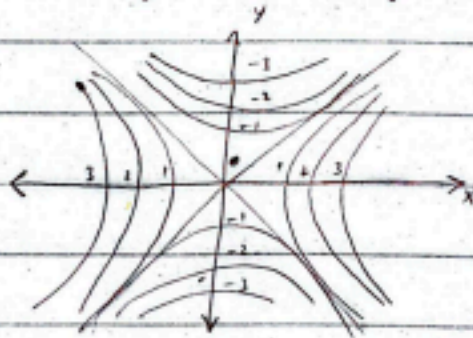
for contour map we take $x^2 - y^2 = c$
Rearranging we get $y^2 = x^2 - c$, which is a
hyperbola's equation.

→ For $c > 0$, The curves open left/right (along x-axis)

→ For $c < 0$, The curves open up/down (along y-axis)

→ For $c = 0$, The curve is the pair of lines $y = \pm x$.

So contour map will show families of hyperbolas
oriented differently depending upon c .



QUESTION : 48

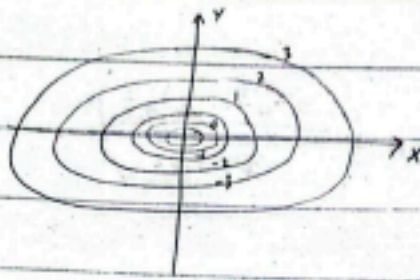
$$f(x, y) = \ln(x^2 + 4y^2)$$

Sol:

for level curves, $\ln(x^2 + 4y^2) = c$

$$x^2 + 4y^2 = e^c \rightarrow \text{equation of ellipse}$$

The contour map will show a family of
concentric ellipses, elongated along x-axis



EXERCISE : 14.2

QUESTION : 5

$$\lim_{(x,y) \rightarrow (3,2)} (x^2y^2 - 4y^2)$$

Sol:

$$\lim_{(x,y) \rightarrow (3,2)} (x^2y^2 - 4y^2) = \text{apply limit ;}$$

$$= [(3)^2(2)^2 - 4(2)^2] = [(9)(8) - 4(4)]$$

$$= 56 \quad \text{Answer.}$$

20/10/20

QUESTION : 7

$$f(x,y) = \frac{x^2y - xy^3}{x - y + 2}$$

$$(x,y) \rightarrow (-3,1)$$

Sol:

Apply limit ,

$$= \frac{(-3)^2(1) - (-3)(1)^3}{-3 - 1 + 2} = \frac{9 + 3}{-4 + 2} = \frac{12}{-2}$$

$$= -6 \quad \text{Answer.}$$























